

Appendix – The Dangers of Non–Self-Fixed Architecture Technical Debt and Its Impact on Time-to-Fix

Edi Sutoyo, Paris Avgeriou, Andrea Capiluppi

Supplementary Material

1 Robustness Analysis Using True-ATD Items Only

This appendix reports a robustness analysis restricted exclusively to issues classified as *True-ATD*. Issues classified as *Weak-ATD* are excluded from all analyses reported below. The purpose of this analysis is to assess whether the main findings remain consistent when only ATD items with stronger classification agreement are considered.

1.1 RQ1—Prevalence of self-fixing in ATD repayment

Figure 1 presents the overall distribution of self-fixed and non-self-fixed items within the True-ATD-only dataset. Of the 577 True-ATD items considered in this robustness analysis, 104 (18.02%) were self-fixed, whereas 473 (81.98%) were fixed by a developer other than the introducer. Thus, approximately one in five True-ATD items was eventually repaid by the same developer who introduced it, while the large majority were repaid by another developer.

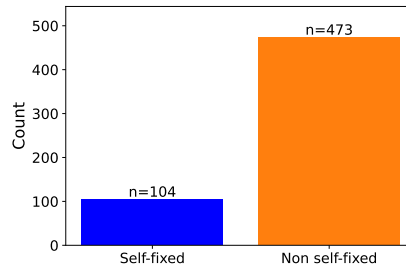


Figure 1: Distribution of self-fixed and non-self-fixed items in the True-ATD-only dataset.

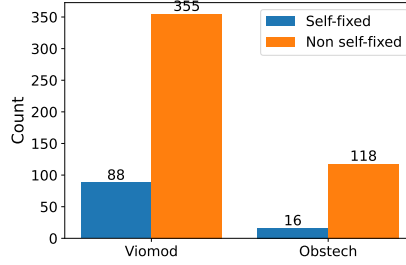


Figure 2: Distribution of self-fixed and non-self-fixed True-ATD items grouped by indicator.

Figure 2 further breaks down the True-ATD-only distribution according to the ATD indicator. Among the 443 *VioMod* items, 88 (19.86%) were self-fixed and 355 (80.14%) were non-self-fixed. For the 134 *ObsTech* items, 16 (11.94%) were self-fixed and 118 (88.06%) were non-self-fixed. Self-fixing is therefore observed for both ATD indicators, although descriptively it occurs more frequently among *VioMod* items than among *ObsTech* items.

Overall, the True-ATD-only analysis shows that self-fixing remains a minority repayment pattern. Most True-ATD items are repaid by developers other than those who introduced them, indicating that the prevalence pattern observed in the broader analysis is preserved when weak-ATD items are excluded.

1.2 RQ1.1—Distribution of commit shares on ATD-affected files

Figure 5 shows the per-file contribution-share distributions for self-fixed items in the True-ATD-only dataset. When contribution is measured using commit counts, the Self-fixer accounts for a larger share of activity on the affected files than the aggregated group of all other developers. The median commit share of the Self-fixer is above 0.5, whereas the median share of Others is below 0.5. The corresponding distributions are complementary because the two contribution shares sum to one for each file.

A similar pattern is observed when contribution magnitude is measured using changed lines. As shown in Figure 5(b), the Self-fixer again has a median contribution share above 0.5, while Others account for a smaller median proportion of the total changed lines. The distribution is relatively wide, however, indicating that the degree of Self-fixer involvement differs considerably across files. Nevertheless, the results from both measures show that, among self-fixed True-ATD items, the developer who eventually repays the debt tends to account for a substantial portion of the contribution activity on the affected files during the ATD lifetime.

For non-self-fixed True-ATD items, the contribution pattern is more distributed across developer roles. Figure 6(a) shows that both the Introducer and the eventual Fixer account for relatively modest shares of the commits made

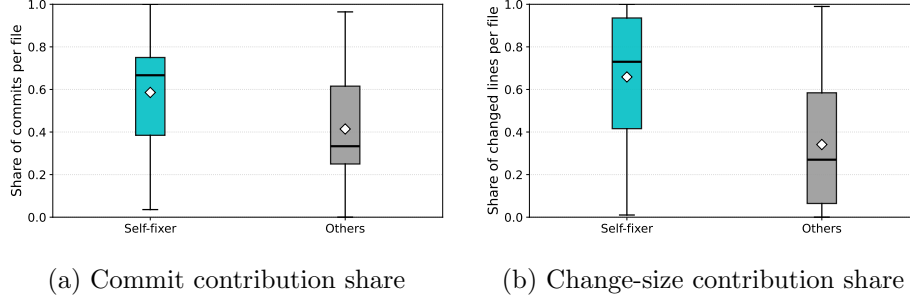


Figure 3: Boxplots of per-file contribution shares for the Self-fixer and Others in self-fixed True-ATD items, based on commit counts and change size.

to the affected files, with their median shares lying below 0.5. In contrast, the aggregated group of Other developers exhibits the largest median commit share. This indicates that commit activity on files affected by non-self-fixed True-ATD is generally not concentrated in either the Introducer or the eventual Fixer alone.

The change-size distributions in Figure 6(b) show a comparable pattern. The Introducer and Fixer have lower median shares of changed lines than the aggregated Others group, although all three distributions exhibit substantial variation across files. The broad ranges indicate that some affected files receive a relatively large proportion of changes from one of the focal developers, whereas others receive most of their changes from developers outside the Introducer–Fixer pair.

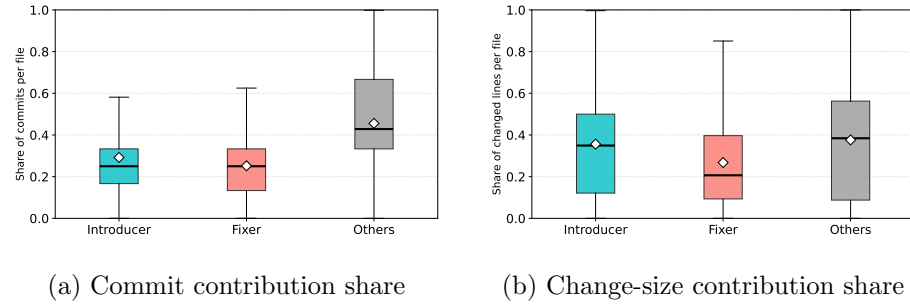


Figure 4: Boxplots of per-file contribution shares for the Introducer, eventual Fixer, and Others in non-self-fixed True-ATD items.

Taken together, the True-ATD-only contribution analysis preserves the same descriptive pattern observed in the broader analysis. In self-fixed True-ATD, contribution activity tends to be concentrated around the Self-fixer, both in terms of commit counts and change magnitude. In non-self-fixed True-ATD, contribution activity is more dispersed, with Other developers collectively ac-

counting for the largest share of activity on the affected files. These results indicate that the observed contribution patterns are not attributable to the inclusion of weak-ATD items.

1.3 RQ1.1—Distribution of commit shares on ATD-affected files

Figure 5 shows the per-file contribution-share distributions for self-fixed ATD. When contribution is measured using commit counts, the Self-fixer accounts for a larger share of activity on the affected files than the aggregated group of all other developers. The median commit share of the Self-fixer is above 0.5, whereas the median share of Others is below 0.5. The corresponding distributions are complementary, because the two contribution shares sum to one for each file.

A similar pattern is observed when contribution magnitude is measured using changed lines. As shown in Figure 5(b), the Self-fixer again has a median contribution share above 0.5, while Others account for a smaller median proportion of the total changed lines. The distribution is relatively wide, however, indicating that the degree of Self-fixer involvement differs considerably across files. Nevertheless, the results from both measures consistently show that, in self-fixed ATD, the developer who eventually repays the debt tends to account for a substantial portion of the contribution activity on the affected files during the ATD lifetime.

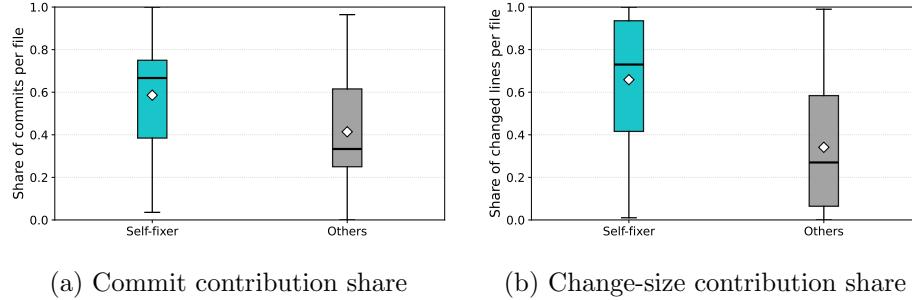


Figure 5: Boxplots of per-file contribution shares for the Self-fixer and Others in self-fixed ATD, based on commit counts and change size.

For non-self-fixed ATD, the contribution pattern is more distributed across developer roles. Figure 6(a) shows that both the Introducer and the eventual Fixer account for relatively modest shares of the commits made to the affected files, with their median shares lying well below 0.5. In contrast, the aggregated group of Other developers exhibits the largest median commit share. This indicates that commit activity on files affected by non-self-fixed ATD is generally not concentrated in either the Introducer or the eventual Fixer alone.

The change-size distributions in Figure 6(b) show a comparable pattern. The Introducer and Fixer have lower median shares of changed lines than the

aggregated Others group, although all three distributions exhibit substantial variation across files. In particular, the broad ranges indicate that some affected files are dominated by one of the focal developers, whereas others receive most of their changes from developers outside the Introducer–Fixer pair.

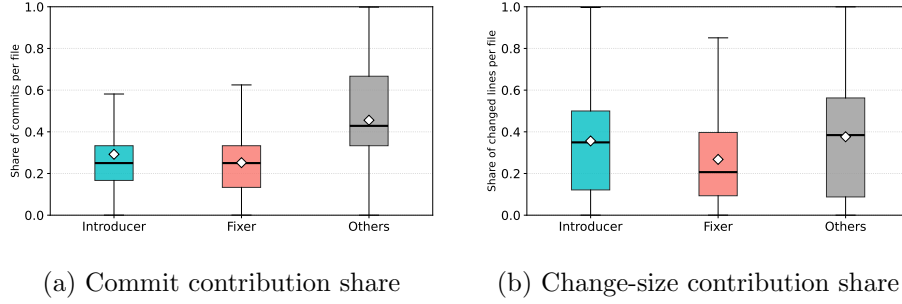


Figure 6: Boxplots of per-file contribution shares for the Introducer, Fixer, and Others in non-self-fixed ATD.

Taken together, the two contribution measures reveal a clear descriptive difference between self-fixed and non-self-fixed ATD. In self-fixed ATD, contribution activity tends to be concentrated around the Self-fixer, both in terms of the number of commits and the magnitude of code changes. In non-self-fixed ATD, contribution activity is more dispersed, with Other developers collectively accounting for the largest share of activity on the affected files.

1.4 RQ2—Time-to-fix comparison between self-fixed and non-self-fixed ATD

As a robustness analysis, we repeated the time-to-fix analysis using only issues classified as *True-ATD*. Issues classified as weak-ATD or non-ATD were excluded from this analysis. This allows us to examine whether the observed difference in repayment time between self-fixed and non-self-fixed ATD remains when the analysis is restricted to cases for which the ATD classification reached a stronger level of agreement.

Figure 7 presents the cumulative fixed curves for the True-ATD-only dataset, comprising 104 self-fixed and 473 non-self-fixed ATD items. The curve for self-fixed ATD rises substantially more rapidly than that for non-self-fixed ATD, particularly during the earlier part of the observation period. This indicates that a larger proportion of self-fixed True-ATD items is repaid within shorter periods after introduction.

The separation between the two curves remains visible over a substantial portion of the observation period. The self-fixed curve approaches its upper range considerably earlier, whereas the cumulative proportion of fixed non-self-fixed items increases more gradually over time. Thus, when the analysis is restricted exclusively to True-ATD items, the descriptive pattern continues to

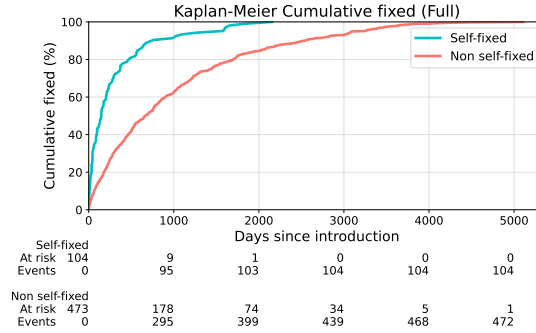


Figure 7: Kaplan–Meier cumulative fixed curves comparing self-fixed and non-self-fixed True-ATD items across the entire observation period.

show shorter repayment times for self-fixed ATD than for non-self-fixed ATD.

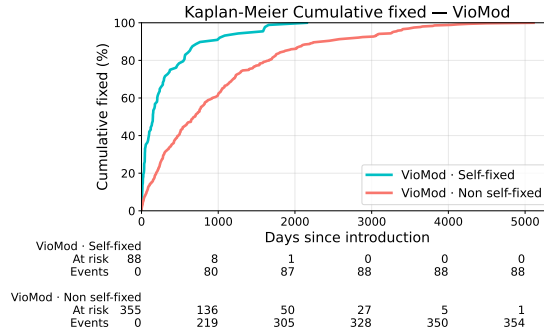


Figure 8: Kaplan–Meier cumulative fixed curves for self-fixed and non-self-fixed VioMod-related True-ATD items.

We further examined whether this pattern is preserved across the two ATD indicators. Figure 8 shows the results for VioMod-related True-ATD items, consisting of 88 self-fixed and 355 non-self-fixed items. The self-fixed curve increases sharply during the earlier stages of the ATD lifetime and remains above the non-self-fixed curve. In contrast, VioMod items repaid by developers other than their introducers accumulate fixes more gradually over time.

The VioMod-specific analysis therefore exhibits the same descriptive pattern as the overall True-ATD analysis: self-fixed items tend to reach repayment earlier than non-self-fixed items.

A similar pattern is observed for ObsTech-related True-ATD items (Figure 9), although this subgroup contains fewer self-fixed observations: 16 self-fixed compared with 118 non-self-fixed items. The cumulative fixed curve for self-fixed ObsTech items rises rapidly and reaches a high cumulative proportion within a relatively short period after introduction. By comparison, the non-

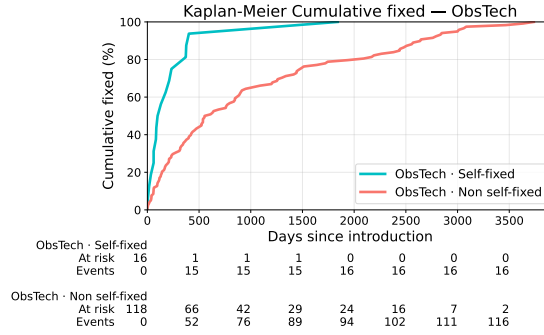


Figure 9: Kaplan–Meier cumulative fixed curves for self-fixed and non-self-fixed ObsTech-related True-ATD items.

self-fixed curve increases considerably more gradually and extends over a much longer observation period.

Taken together, the True-ATD-only robustness analysis yields a consistent descriptive pattern across the overall dataset and both ATD indicators. Restricting the analysis to issues with stronger evidence of ATD does not alter the main observation: self-fixed ATD items tend to be repaid earlier than non-self-fixed ATD items. This provides additional evidence that the observed time-to-fix pattern is not driven by the inclusion of weak-ATD items.